

Assignment 19.2 Ai Assisted Coding

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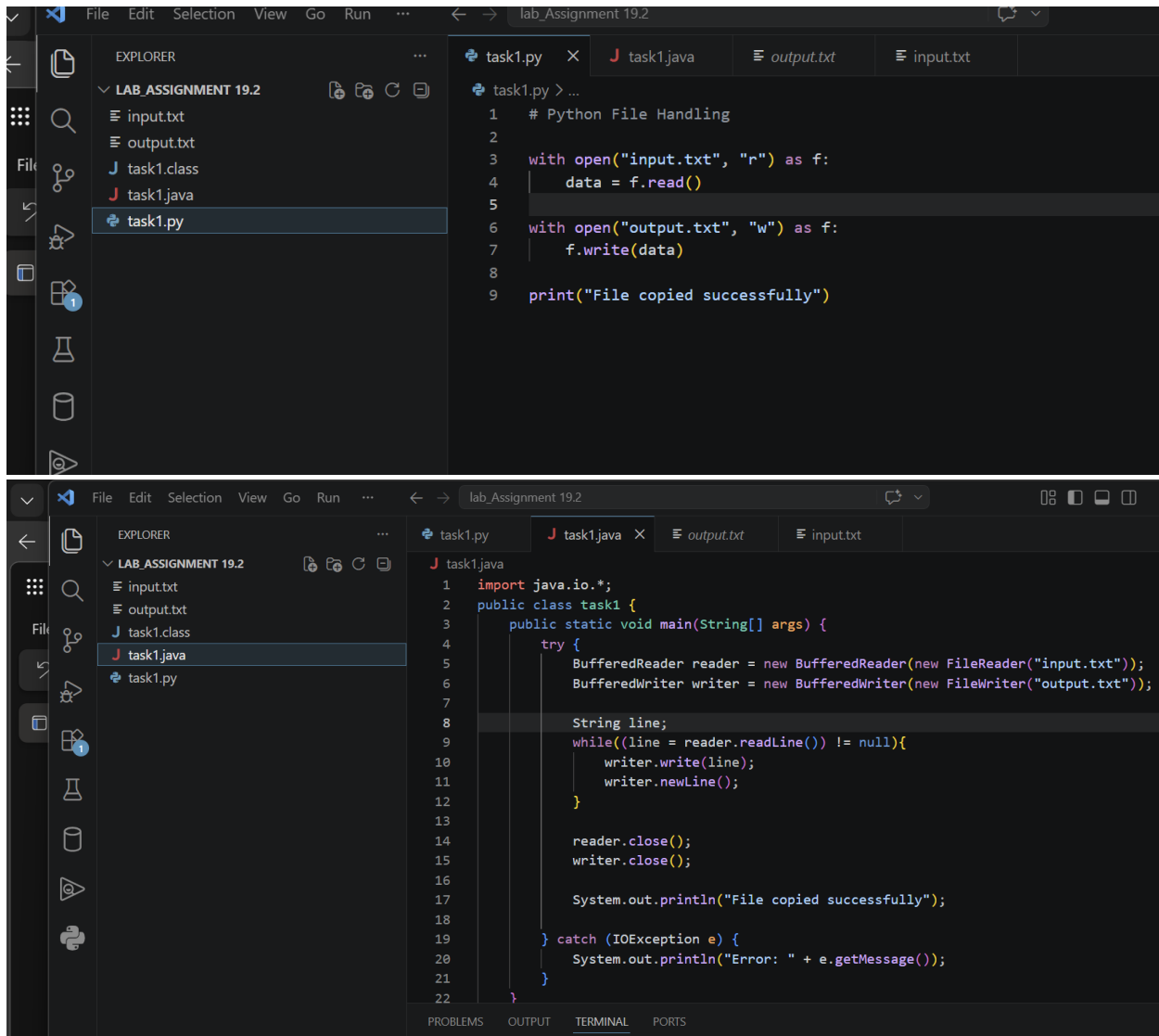
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Task 1: Python to Java Conversion (File Handling)

Prompt:

Convert a Python file handling program into Java.

Code:



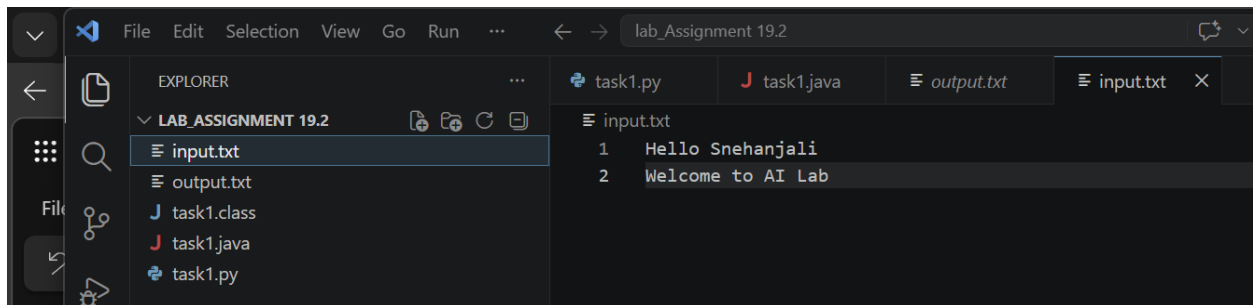
The image displays two screenshots of a code editor interface, likely Visual Studio Code, showing the conversion of a Python file handling program into Java.

Top Screenshot: The editor shows the Python file `task1.py` open. The code in `task1.py` is as follows:

```
1 # Python File Handling
2
3 with open("input.txt", "r") as f:
4     data = f.read()
5
6 with open("output.txt", "w") as f:
7     f.write(data)
8
9 print("File copied successfully")
```

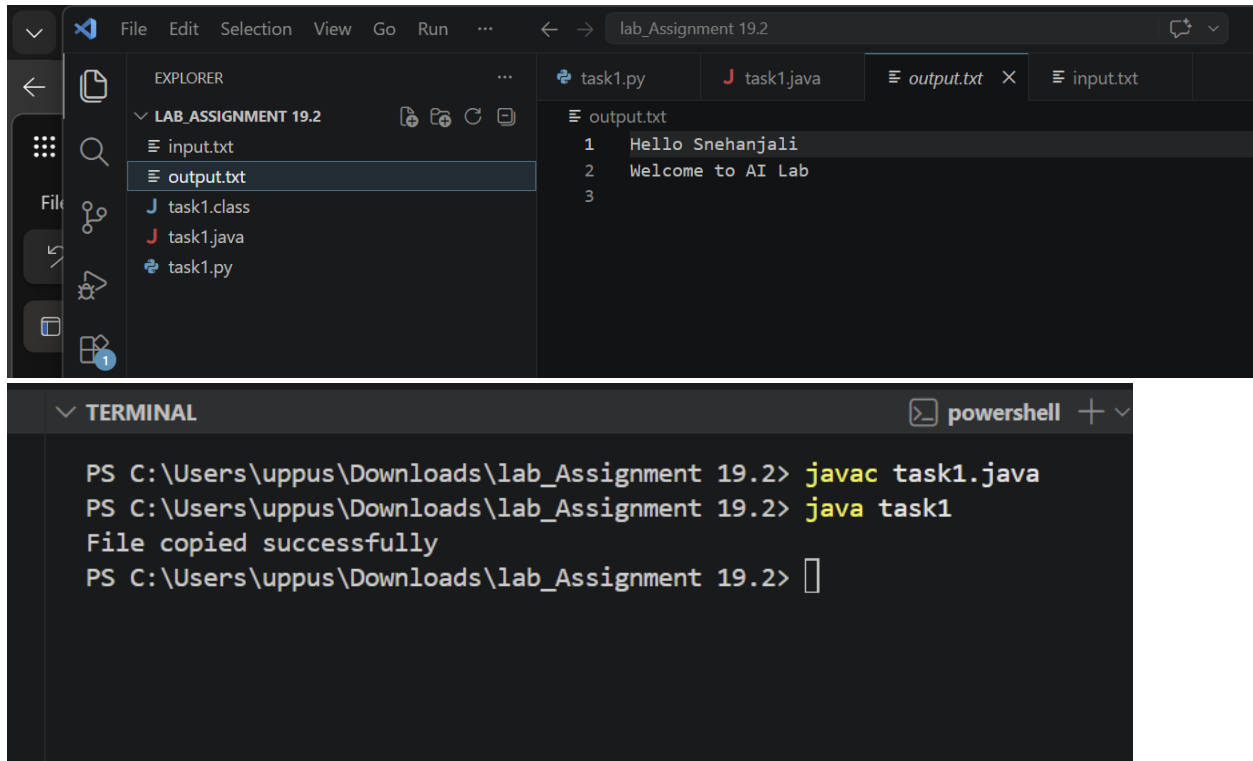
Bottom Screenshot: The editor shows the Java file `task1.java` open. The code in `task1.java` is as follows:

```
1 import java.io.*;
2 public class task1 {
3     public static void main(String[] args) {
4         try {
5             BufferedReader reader = new BufferedReader(new FileReader("input.txt"));
6             BufferedWriter writer = new BufferedWriter(new FileWriter("output.txt"));
7
8             String line;
9             while((line = reader.readLine()) != null){
10                 writer.write(line);
11                 writer.newLine();
12             }
13
14             reader.close();
15             writer.close();
16
17             System.out.println("File copied successfully");
18
19         } catch (IOException e) {
20             System.out.println("Error: " + e.getMessage());
21         }
22     }
23 }
```



```
File Edit Selection View Go Run ... lab_Assignment 19.2
EXPLORER
LAB_ASSIGNMENT 19.2
input.txt
output.txt
task1.class
task1.java
task1.py
input.txt
1 Hello Snehanjali
2 Welcome to AI Lab
```

Output:



```
File Edit Selection View Go Run ... lab_Assignment 19.2
EXPLORER
LAB_ASSIGNMENT 19.2
input.txt
output.txt
task1.class
task1.java
task1.py
output.txt
1 Hello Snehanjali
2 Welcome to AI Lab
3
TERMINAL powershell
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> javac task1.java
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> java task1
File copied successfully
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

The Python program reads data from a file and writes it to another file using open() function.

The Java program uses BufferedReader and BufferedWriter for file handling.

Exception handling is required in Java.

Both programs perform the same operation and produce identical output.

Task2: C++ to Python Conversion (Sorting)

Prompt:

Convert a C++ sorting program into Python.

Code:

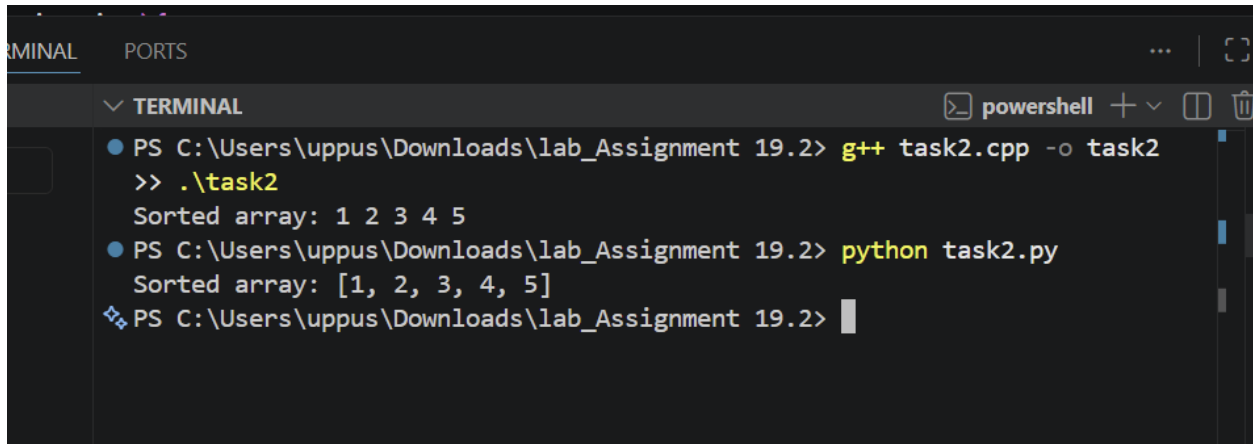
The screenshot shows the Visual Studio Code interface with the file explorer on the left displaying the contents of 'LAB_ASSIGNMENT 19.2'. The files listed are input.txt, output.txt, task1.class, task1.java, task1.py, task2.cpp (selected), task2.exe, and task2.py. The main editor window shows the code for task2.cpp, which implements a bubble sort algorithm. The code includes iostream and namespace std, defines an array arr of size 5 with values {5, 2, 4, 1, 3}, and uses nested loops to sort the array. The output is printed as 'Sorted array: ' followed by the elements of the array.

```
task2.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int arr[5] = {5, 2, 4, 1, 3};
6
7      for(int i=0;i<5;i++){
8          for(int j=0;j<5-i-1;j++){
9              if(arr[j] > arr[j+1]){
10                 swap(arr[j], arr[j+1]);
11             }
12         }
13     }
14
15     cout << "Sorted array: ";
16     for(int i=0;i<5;i++)
17         cout << arr[i] << " ";
18
19     return 0;
20 }
```

The screenshot shows the Visual Studio Code interface with the file explorer on the left displaying the contents of 'LAB_ASSIGNMENT 19.2'. The files listed are input.txt, output.txt, task1.class, task1.java, task1.py, task2.cpp, task2.exe, and task2.py (selected). The main editor window shows the code for task2.py, which implements a bubble sort algorithm. The code defines an array arr with values [5, 2, 4, 1, 3] and uses nested loops to sort the array. The output is printed as 'Sorted array:' followed by the elements of the array.

```
task2.py > ...
1  arr = [5, 2, 4, 1, 3]
2
3  for i in range(len(arr)):
4      for j in range(len(arr) - i - 1):
5          if arr[j] > arr[j + 1]:
6              temp = arr[j]
7              arr[j] = arr[j + 1]
8              arr[j + 1] = temp
9
10 print("Sorted array:", arr)
```

Output:



```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> g++ task2.cpp -o task2
>> .\task2
Sorted array: 1 2 3 4 5
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task2.py
Sorted array: [1, 2, 3, 4, 5]
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

-->The C++ program uses Bubble Sort to sort an array.

-->The Python program follows the same logic.

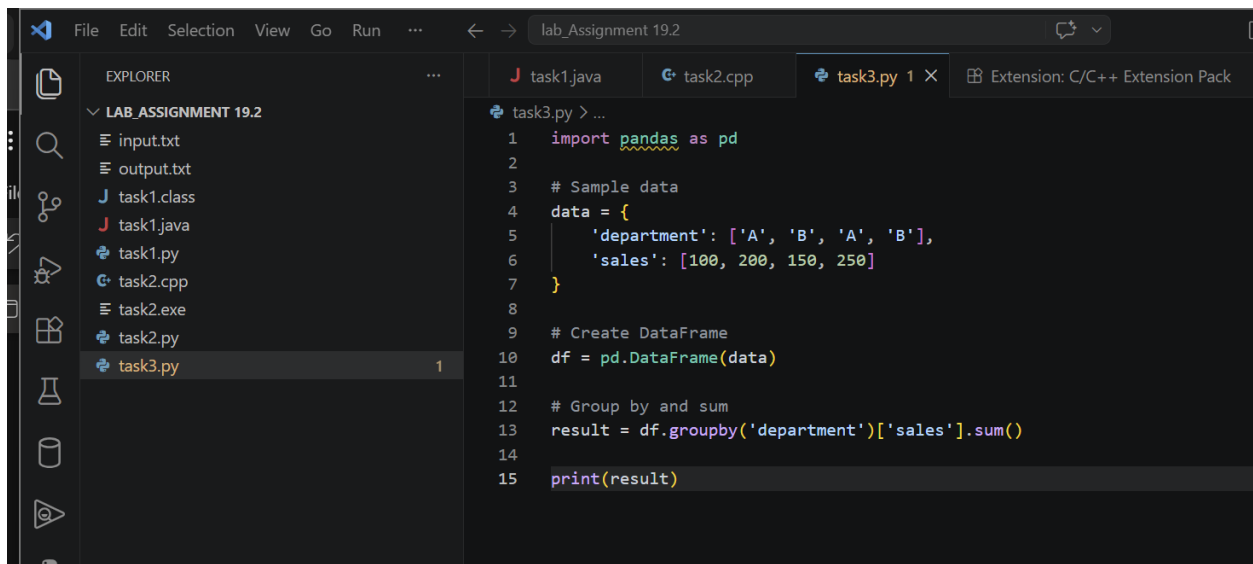
-->Both programs produce the same sorted result.

Task 3: SQL to Pandas Conversion

Prompt:

Convert a SQL query with GROUP BY into Pandas.

Code:



```
1 import pandas as pd
2
3 # Sample data
4 data = {
5     'department': ['A', 'B', 'A', 'B'],
6     'sales': [100, 200, 150, 250]
7 }
8
9 # Create DataFrame
10 df = pd.DataFrame(data)
11
12 # Group by and sum
13 result = df.groupby('department')['sales'].sum()
14
15 print(result)
```

Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task3.py
department
A      250
B      450
Name: sales, dtype: int64
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

-->The SQL query groups data by department and calculates total sales.

-->The Pandas code uses `groupby()` and `sum()` to achieve the same result.

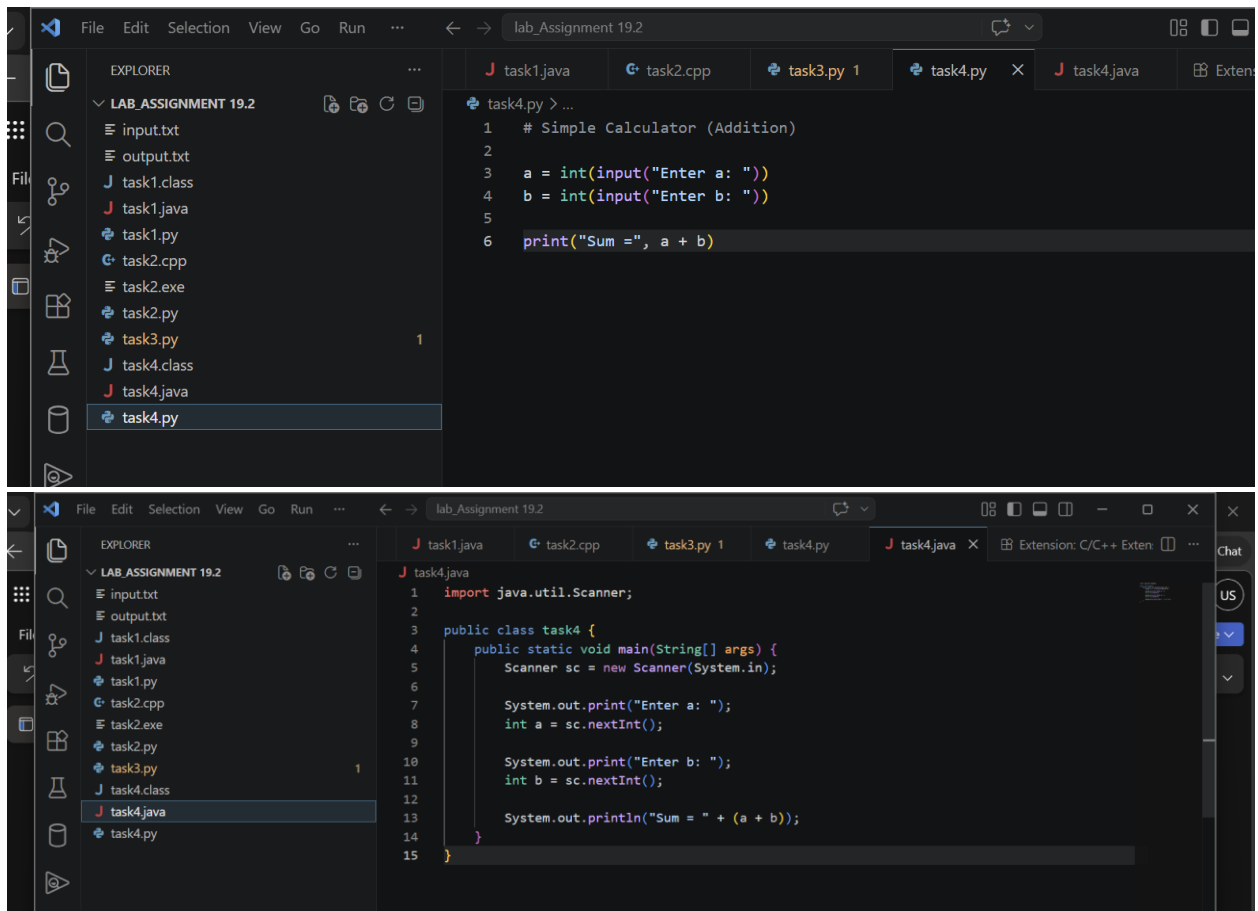
-->Both produce identical outputs.

Task 4: Real-Time Code Conversion (Python to Java)

Prompt:

Convert a Python calculator program into Java.

Code:



Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task4.py
Enter a: 3
Enter b: 5
Sum = 8
```

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> javac task4.java
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> java task4
Enter a: 3
Enter b: 5
Sum = 8
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> 
```

Explanation:

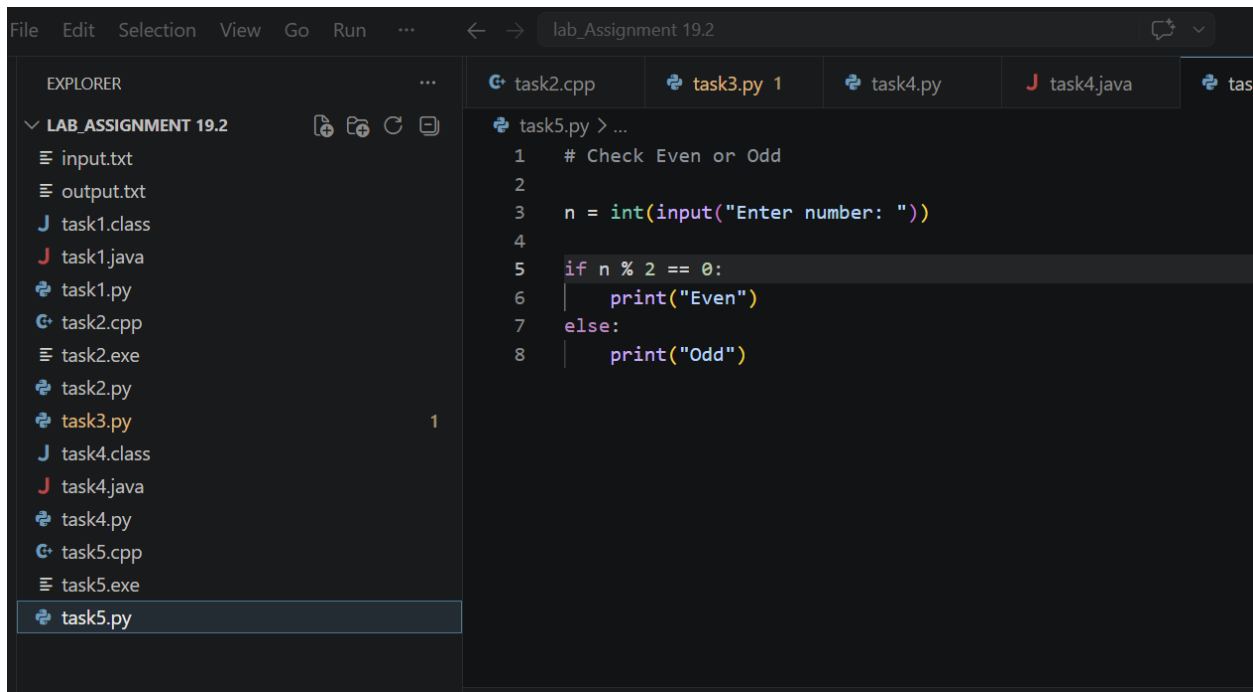
- >The Python program performs addition using simple input and print statements.
- >The Java program uses Scanner for input and follows class-based structure.
- >Both programs perform the same operation and produce identical output.

Task5: Python to C++ Conversion (Even/Odd)

Prompt:

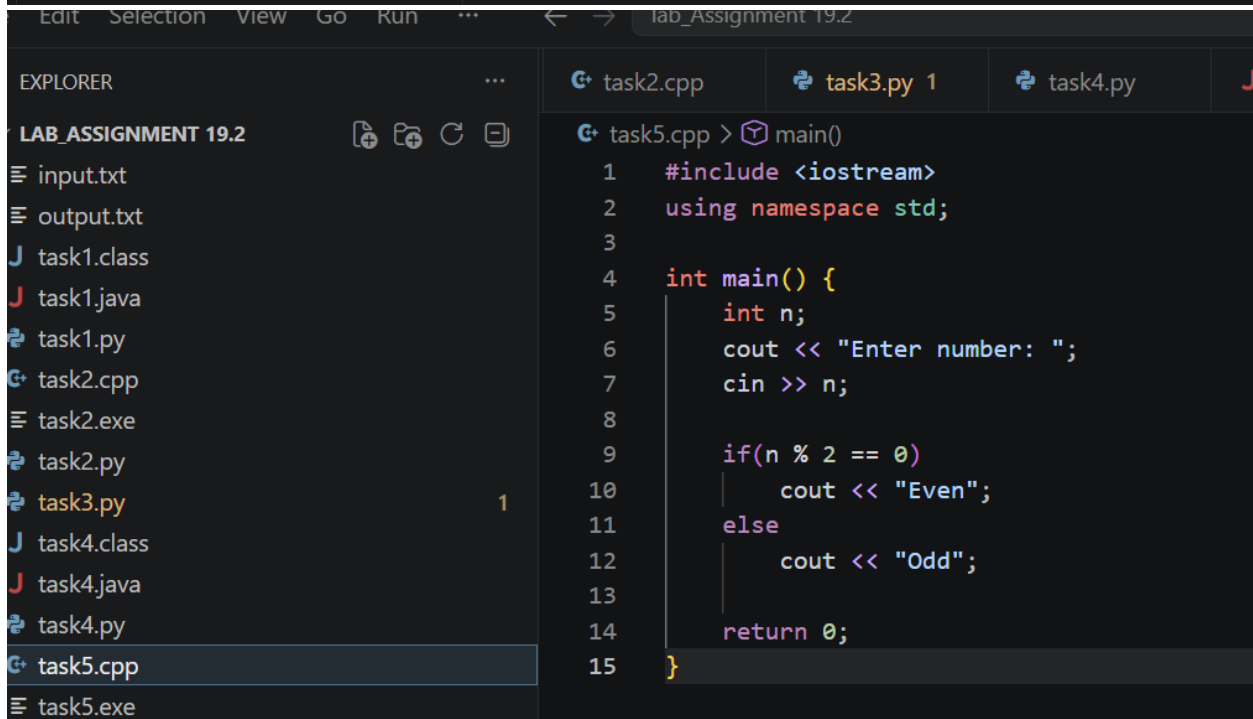
Convert a Python program to C++ to check even or odd number.

Code:



The screenshot shows the Visual Studio Code editor with the file explorer on the left and the code editor on the right. The file explorer shows a project named 'LAB_ASSIGNMENT 19.2' with various files including 'task5.py'. The code editor displays the Python code for checking even or odd numbers.

```
1 # Check Even or Odd
2
3 n = int(input("Enter number: "))
4
5 if n % 2 == 0:
6     print("Even")
7 else:
8     print("Odd")
```



The screenshot shows the Visual Studio Code editor with the file explorer on the left and the code editor on the right. The file explorer shows a project named 'LAB_ASSIGNMENT 19.2' with various files including 'task5.cpp'. The code editor displays the C++ code for checking even or odd numbers.

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n;
6     cout << "Enter number: ";
7     cin >> n;
8
9     if(n % 2 == 0)
10         cout << "Even";
11     else
12         cout << "Odd";
13
14     return 0;
15 }
```

Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task5.py
Enter number: 3
Odd
● PS C:\Users\uppus\Downloads\lab_Assignment 19.2> g++ task5.cpp -o task5
>> .\task5
Enter number: 3
○ Odd
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> 
```

Explanation:

-->The Python program checks whether a number is even or odd using modulus operator.

-->The C++ program follows the same logic using if-else condition.

-->Both programs produce correct results for all test cases.